

Recovery Stress and Recovery Strain of TiNi Shape Memory Alloy*

(Cyclic Properties under Constant Residual Strain and Constant Maximum Stress)

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The cyclic behavior of recovery stress under constant residual strain and recovery strain under constant maximum stress in TiNi shape memory alloy was investigated experimentally. The influence of residual strain, maximum stress and the heating temperature on cyclic characteristics of recovery stress and recovery strain was discussed. The main results are summarized as follows. (1) The stress at high temperature, the stress at low temperature and the recovery stress decrease in the early cycles, but become almost constant after these cycles. (2) The recovery stress under thermal cycling decreases with residual strain. (3) Both the recovery stress and the stress at high temperature increase with the heating temperature, but the stresses' rates of increase decrease. (4) The transformation temperature under constant stress rises with the number of thermal cycles.

Key Words: Inelastic Deformation, Nonferrous Material, Titanium-Nickel Shape Memory Alloy, Recovery Stress, Recovery Strain, Thermal Cycling

1. Introduction

After the removal of the load for shape memory alloy at temperatures below the reverse-transformation finish temperature (A_f), there appears a certain amount of residual strain. This residual strain is caused by the stress-induced martensitic (SIM) transformation and disappears upon heating above A_f under no load. This phenomenon in which the material recovers its original form by heating is called the shape memory effect^{(1),(2)}. If the material is heated above A_f with the residual strain appearing after the removal of the load maintained constant, recovery stress which is larger than the stress appearing in the loading process may occur^{(3),(4)}. This recovery stress is

used in many applications, such as in the industrial field and in the medical field^{(5),(6)}. From the standpoint of evaluating the reliability in the mechanical properties of shape memory alloy as the functional material in an application, the behaviors of transformation temperature, recovery deformation and recovery stress under the thermomechanical cycling are very important problems⁽⁷⁾. For these purposes, the authors have studied the cyclic properties on the shape memory effect⁽⁸⁾, pseudoelasticity^{(9),(10)}, recovery stress⁽¹¹⁾ and recovery strain⁽¹²⁾ of TiNi shape memory alloy.

In the current paper, following the previous work on recovery stress under constant maximum strain⁽¹¹⁾, the cyclic recovery stress under constant residual strain and cyclic recovery strain under constant maximum stress will be discussed. The recovery stress under constant residual strain has been used in many applications for the shape memory alloy elements, for example, the tightening force in a coupling^{(5),(6)}. If the element is applied for a pipe coupling, the variation of the recovery stress caused by the temperature variation of the fluid is an important problem. Nevertheless, the cyclic properties of the recovery stress have been reported on by a few

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researchers. The factors which affect the recovery stress and recovery strain are shape memory processing temperature, residual strain, maximum strain, heating temperature and cooling temperature.

In the present study, concerning maximum strain and heating temperature, the cyclic behavior of recovery stress under constant residual strain and recovery strain under constant maximum stress in the thermal cycling of TiNi shape memory alloy was investigated experimentally.

2. Experimental Procedures

2.1 Material and specimen

The material was a Ti-55.3 wt%Ni alloy wire produced by Furukawa Electric Co., Ltd. The specimen was shape-memorized into a straight line from the wire. In the shape memory processing, the wire was kept at 673K for 20 min and cooled in a furnace. The temperature A_f was about 323 K. The outer diameter of the specimen was 0.75 mm and the gauge length was 20 mm.

2.2 Experimental apparatus

The experimental apparatus consists of a tensile machine and a heating-cooling device⁽⁹⁾. Axial force, displacement and temperature were measured by a load cell, an extensometer and a thermocouple, respectively. The specimen was heated and cooled by hot air and cold air.

2.3 Experimental tests

(a) Experiment A on recovery stress

The schematic stress-strain relation in the experiments A and B is shown in Fig. 1. In Exp. A on recovery stress, the maximum strain ϵ_m was given at the temperature T_1 below A_f by tension at first, and the residual strain ϵ_0 resulted from unloading from ϵ_m . With the strain ϵ_0 kept constant, the specimen was heated at the temperature T_2 above A_f and then stress

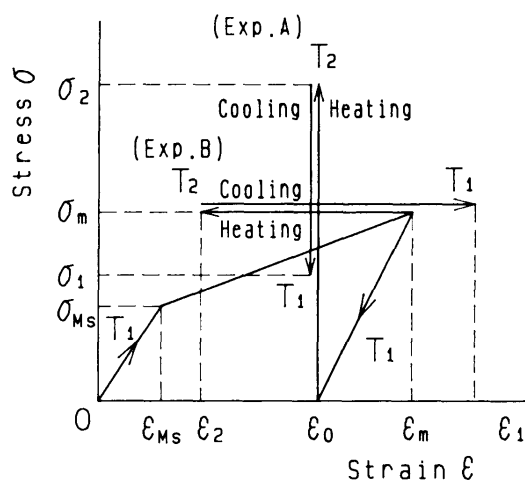


Fig. 1 The schematic stress-strain relation in Exps. A and B

was increased to σ_2 . Under this process, if the specimen was cooled to T_1 under constant ϵ_0 , stress decreased to σ_1 . These heating and cooling processes completed one cycle and were repeated till the 100th cycle.

(b) Experiment B on recovery strain

At first, the initial strain ϵ_m with the corresponding stress σ_m was given at the temperature T_1 . Following this process and keeping the stress σ_m constant, we repeated the process with heating at T_2 followed by cooling at T_1 100 times.

In Exps. A and B, the values of ϵ_m were 1, 2, 4 and 6%. The value of T_1 was 293 K and the values of T_2 were 333 K, 353 K and 373 K. The rate of heating and cooling was about 1.5 K/s. The strain rate till ϵ_m was 2 %/min.

3. Experimental Results and Discussion

3.1 Recovery stress

3.1.1 Stress-strain curve The stress-strain curves obtained by Exp. A on recovery stress for maximum strain $\epsilon_m=1, 2, 4$ and 6% are shown simultaneously for high temperature $T_2=373$ K in Fig. 2. In Fig. 2, for each ϵ_m , variations in strain for the thermal cycles $N=1$ through 100 are drawn one upon another on the ordinate. In the figure, stress at high temperature σ_2 and stress at low temperature σ_1 in $N=1$ and $N=100$ are indicated with several symbols. Also in the figure, a stress-strain curve obtained by a tensile test at T_2 for maximum strain of 8% is shown with a thin curve. As found from the curve in an initial loading process at T_1 , maximum strain $\epsilon_m=1\%$ is near the starting point of the SIM transformation, ϵ_m

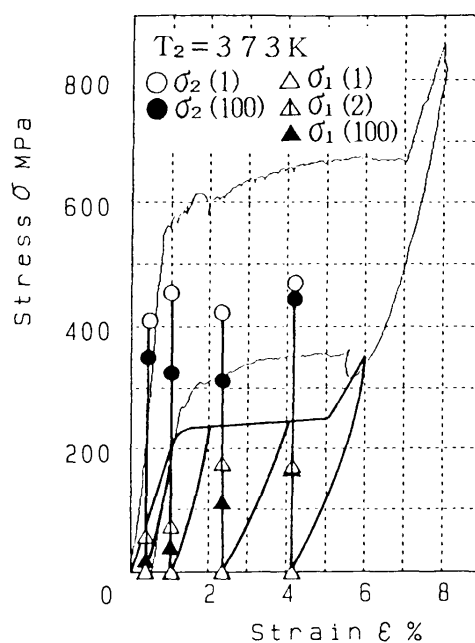


Fig. 2 The stress-strain curves obtained by Exp. A on recovery stress ($T_2=373$ K, $\epsilon_m=1, 2, 4$ and 6%)

=2% and 4% are in the SIM transformation region, and $\varepsilon_m=6\%$ is in the hardening region of the martensitic phase.

The phenomena observed in each test are summarized as follows. (1) The stress at high temperature σ_2 takes almost the same value at each T_2 . (2) The stress σ_2 which arises in the first heating process is lower by about 150MPa than the stress on the loading curve at T_2 and higher by about 100MPa than the stress on the unloading curve at T_2 . (3) The stress at low temperature σ_1 increases with an increase in residual strain ε_0 . (4) The stress σ_1 which arises in the first cooling process is lower by about 100MPa than the stress on the loading curve at the cooling temperature T_1 . (5) Both σ_2 and σ_1 decrease significantly in the early cycles and the rate of these variations decreases with an increase in N . (6) The recovery stress σ_r defined by the difference between σ_2 and σ_1 decreases in proportion to ε_m .

3.1.2 Stress behavior with heating and cooling

In order to investigate stress behavior with temperature variation in each cycle, for maximum strain $\varepsilon_m=4\%$ and the heating temperature $T_2=373\text{ K}$, the relationship between stress and temperature in the number of cycles $N=1$ and 100 is shown in Fig. 3. In the figure, the starting stress and the completing stress of the SIM transformation and the reverse transformation σ_{Ms} , σ_{Mf} , σ_{As} and σ_{Af} , which were obtained by the tensile test under the constant temperature, are plotted with several symbols, which are connected by the dashed curves. These dashed curves represent the starting line and the completing line of the SIM transformation and the reverse transformation⁽³⁾.

As seen in Fig. 3, in the first loading process at $T_1=293\text{ K}$ the initial stress σ_m arises corresponding to the maximum strain ε_m , and in the unloading process the residual strain ε_0 appears. Following these processes, in the heating process under the constant residual strain ε_0 , stress begins to increase from about 300K, increases almost linearly with a slightly larger gradient than that of the starting line of the reverse trans-

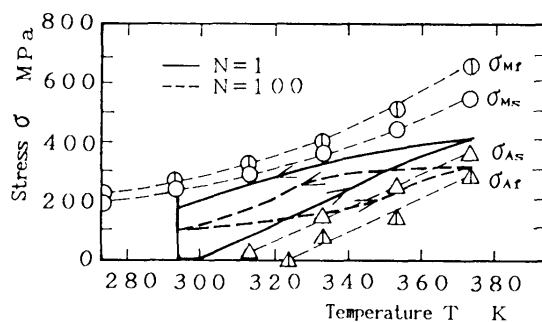


Fig. 3 Relationship between stress and temperature for $N=1$ and 100 ($\varepsilon_m=4\%$ and $T_2=373\text{ K}$)

formation σ_{As} , and finally reaches σ_2 at T_2 . Following this process, in the cooling process under the constant ε_m , stress decreases along the curve nearly parallel to the starting line of the SIM transformation σ_{Ms} with a slightly smaller gradient than that of the curve in the heating process and finally reaches σ_1 at T_1 .

In the theoretical analysis⁽⁹⁾, in the heating process, stress increases along the starting line of the reverse transformation σ_{As} . Comparing the stress σ_{As} in Fig. 3, we find that the experimental data are larger than σ_{As} . The reason the predicted stress σ_{As} is smaller than the actual stress may depend on the analysis in which the transformation-strain range is assumed to be constant. In the theory, if the material is heated under the constraint of deformation, recovery stress increases in proportion to the transformation-strain range⁽³⁾. In practice, as observed in Fig. 2, by comparison with the initial loading curves at T_1 and T_2 , the strain range of the SIM transformation increases in proportion to the temperature^{(9),(10)}. Thus, if recovery stress is calculated under the constant strain range, the predicted stress may be smaller than the actual one.

As seen in Fig. 3, the stress level in $N=100$ is lower than that in $N=1$. In the heating process in $N=100$, while stress does not vary till about 320K, stress increases above 320K and the stress-temperature curve exhibits a long and thin S-shaped form. The reason the temperature at which stress starts to increase in the heating process in $N=100$ is higher than that in $N=1$ is based on the fact that, as observed in the cyclic tests on the shape memory effect and pseudoelasticity, the starting line of the reverse transformation moves toward the high-temperature direction with the number of cycles⁽⁸⁾⁻⁽¹⁰⁾. The phenomenon in which the stress-temperature curve shows the S-shaped form has the same inclination as that on the recovery stress under the constant maximum strain⁽¹¹⁾.

3.1.3 Behavior of recovery stress with thermal cycling As shown schematically in Fig. 1, stress increases from σ_1 to σ_2 by heating from T_1 to T_2 . The amount of stress increment $\Delta\sigma = \sigma_2 - \sigma_1$ represents the effective recovery stress σ_r caused by heating. In order to investigate the cyclic behavior of these stresses, stress at high temperature σ_2 , stress at low temperature σ_1 and recovery stress $\sigma_r = \Delta\sigma$ in relation to the number of cycles N in the case of $\varepsilon_m=4\%$ and $T_2=373\text{ K}$ are shown in Fig. 4. In the case of $N=1$, $\sigma_1=0$, and so $\sigma_r = \sigma_2$.

It is seen in Fig. 4 that σ_2 , σ_1 and σ_r decrease in the early cycles and take almost constant values in $N > 30$. The amount of decrease till $N=100$ in σ_2 is about 100 MPa and that in σ_1 and σ_r is about 50 MPa.

In the case of another maximum strain ε_m and heating temperature T_2 , almost the same cyclic behavior of σ_2 , σ_1 and σ_r is observed.

It may be said that, in the application of σ_2 or σ_r , because these stresses take almost constant values except in the early cycles, training⁽¹³⁾ before practical use is effective in improving reliability in the cyclic recovery stress.

3.1.4 Dependence of recovery stress on residual strain In order to investigate dependence of the cyclic behavior of recovery stress on the residual strain ε_0 , concerning the number of cycles $N=2$ and 100 as the representative of the start and the end of thermal cycles, stress at high temperature σ_2 , stress at low temperature σ_1 and recovery stress σ_r in relation to ε_m at heating temperature $T_2=333$ K are shown in Fig. 5. In Fig. 5, because there is no cooling process in $N=1$ and recovery stress by heating is considered, the stresses in $N=1$ are avoided. In the figure, the starting stresses of the SIM transformation and the reverse transformation σ_{Ms} and σ_{As} obtained by the tensile tests at T_2 are plotted.

It is seen in Fig. 5 that, in the range $\varepsilon_0 \geq 1\%$, dependence of σ_2 on ε_0 is slight and σ_2 takes the value near σ_{As} . The stress σ_1 increases with an increase in ε_0 for $\varepsilon_0 \leq 2.5\%$. This phenomenon may be explained by the fact that ε_0 corresponds to the strain caused by the

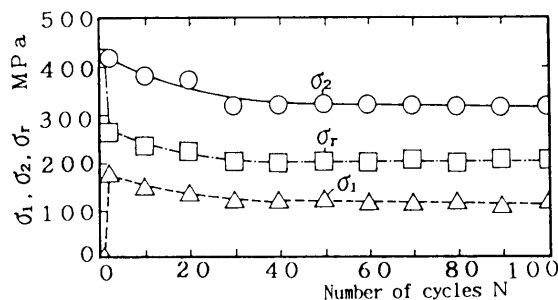


Fig. 4 Changes in stresses σ_2 , σ_1 and σ_r as a function of repetition cycles ($\varepsilon_m=4\%$ and $T_2=373$ K)

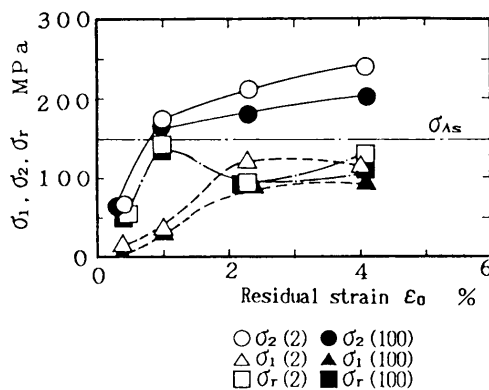


Fig. 5 Dependence of stresses σ_2 , σ_1 and σ_r on residual strain ($T_2=333$ K)

SIM transformation and so the strain of the reverse transformation caused by heating increases in proportion to ε_0 . In the range $1\% \leq \varepsilon_0 \leq 2.5\%$, that is, if the initial maximum strain ε_m is in the range of SIM transformation between 2% and 4%, σ_r decreases significantly with an increase in ε_0 .

From these discussions, it may be said that, in the application under thermal cycling, if the maximum strain ε_m is in the range of the SIM transformation, (1) a small amount of ε_m is effective in obtaining a high level of σ_r and (2) a small amount of ε_m is sufficient to obtain a high level of σ_2 .

3.1.5 Dependence of recovery stress on heating temperature In order to investigate the dependence of the cyclic behavior of recovery stress on the heating temperature T_2 , for the same numbers of cycles, $N=2$ and 100, as in Fig. 5, stress at high temperature σ_2 , stress at low temperature σ_1 and recovery stress σ_r in relation to T_2 in the case of $\varepsilon_m=2\%$ are shown in Fig. 6. In Fig. 6, the starting lines of the SIM transformation and the reverse transformation σ_{Ms} and σ_{As} obtained by the tensile tests at the constant temperature T_2 are plotted.

It is seen in Fig. 6 that all stresses σ_2 , σ_1 and σ_r increase significantly with the rise in T_2 below 353 K but increase slightly for $T_2 \geq 353$ K. The reason why both σ_2 and σ_1 at $T_2=333$ K are small may be considered to be as follows. Because A_f of the material is 323 K and the difference between T_2 and A_f is 10 K, the thermal energy required to cause the reverse transformation is small and so σ_2 increases a little. The reason why σ_2 increases slightly at $T_2 \geq 353$ K may be that the reverse transformation progresses with the rise in T_2 and the stress-temperature curve approaches the completing line σ_{Af} from the starting line of the reverse transformation σ_{As} , meaning that the gradient of the curve decreases with the rise in T_2 ⁽¹⁴⁾.

On the basis of the fact that both σ_2 and σ_r increase slightly in $T_2 \geq 353$ K, it is ascertained that the

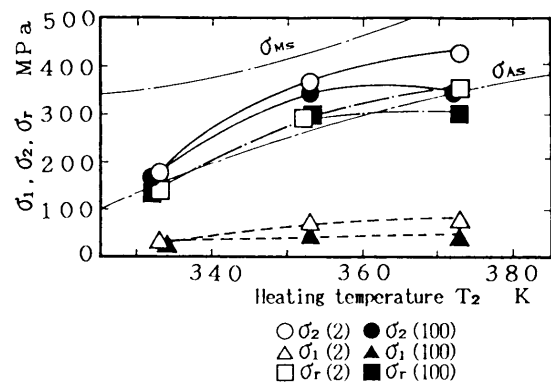


Fig. 6 Dependence of stresses σ_2 , σ_1 and σ_r on heating temperature ($\varepsilon_m=2\%$)

effective heating temperature for obtaining the high level of σ_2 or σ_r is $A_f + 30$ K.

By comparison with each stress in $N=2$ and $N=100$, both σ_2 and σ_1 in $N=100$ are lower than those in $N=2$ and the amount of decrease in these values increases with the rise in T_2 . The values of σ_r in $N=2$ and 100 almost coincide except for at $T_2=373$ K.

In the case of another maximum strain ε_m , dependence of σ_2 , σ_1 and σ_r on T_2 shows almost the same inclination.

3.1.6 Cyclic behavior of stress at high temperature The relationship between stress at high temperature σ_2 and the number of cycles N discussed in Fig. 4 and the relationship between the starting stress of the SIM transformation σ_{Ms} and N obtained in the cyclic deformation on pseudoelasticity at the constant temperature T_2 as discussed in the previous paper⁽¹⁰⁾ in the case of $\varepsilon_m=2\%$ are shown in Fig. 7.

It is seen in Fig. 7 that both σ_{Ms} and σ_2 decrease significantly in the early cycles at every T_2 . While the amounts of decrease in σ_{Ms} for $N=1\sim 100$ are 100~200 MPa, those in σ_2 are less than 100 MPa.

The reason a difference between the amounts of decrease in σ_{Ms} and σ_2 appears may be considered to be as follows. In each cycle of the cyclic deformation on pseudoelasticity, while the transformation strain appears due to the SIM transformation in the loading process, the transformation strain disappears due to the reverse transformation in the unloading process. In the interface between the martensitic phase and the parent phase, the dislocation may appear readily. Corresponding to the loading and unloading processes, growth and disappearance of the martensitic phase by both transformations occur, and the interface moves. Because the interface moves repeatedly with the cyclic loading and unloading processes, and hence the internal stress caused by the dislocation acts repeatedly, the stress σ_{Ms} decreases significantly⁽⁸⁾. On the contrary, in the thermal cycling under the constant residual strain ε_0 , because ε_0 is kept constant, it is hard for the reverse transformation to occur, and the amount of variation in the fraction of the martensitic

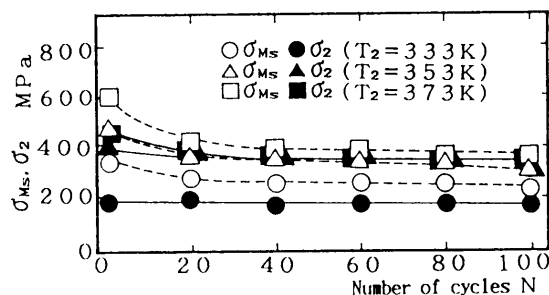


Fig. 7 Changes in stresses σ_2 and σ_{Ms} as a function of repetition cycles ($\varepsilon_m=2\%$)

phase is only 0.08 in the analysis for $T_2=353\text{K}^{(3)}$. Compared with the cyclic deformation on pseudoelasticity, the interface between the martensitic phase and the parent phase moves slightly, and thus, σ_2 decreases slightly.

3.2 Recovery strain

3.2.1 Stress-strain curve The stress-strain curves obtained by Exp. B on recovery strain with the thermal cycling under constant stress are shown in Fig. 8. In Fig. 8, the whole curves for the case of initial strain $\varepsilon_m=1, 2$ and 6% at high temperature $T_2=373$ K are plotted. In the figure, the strain at high temperature ε_2 and the strain at low temperature ε_1 for $N=1$ and $N=100$ and ε_1 for $N=2$ are indicated by several symbols. Also in the figure, a stress-strain curve obtained by a tensile test at 373 K for a maximum strain of 8% is shown with a thin curve. As found from the curve in the initial loading process at T_1 , the initial strain (ε_m) value near the starting point is 1% in the SIM transformation, 2% in the SIM transformation region, and 6% in the hardening region of the martensitic phase.

It is seen in Fig. 8 that the difference in strain between high temperature and low temperature is large for $\varepsilon_m=2\%$ but small for $\varepsilon_m=1\%$ and 6% . If ε_m is in the SIM transformation region as it is for $\varepsilon_m=2\%$, ε_1 for $N=100$ is about 5% , which is nearly equal to the strain of the complete SIM transformation ε_{Mf} at T_1 .

3.2.2 Strain behavior with heating and cooling

We have plotted in Fig. 9, for each initial strain ε_m and high temperature T_2 , the relationship between strain and temperature as it changes with the number of cycles $N=1, 2, 5, 20, 50$ and 100 .

The phenomena observed in each experiment may be summarized as follows. (1) Although $T_2 > A_f$, if T_2 is not higher than a certain temperature, recovery strain by reverse transformation does not appear. (2) Both starting temperature and completing temperature of the SIM transformation and the reverse transformation rise with N , and the position of the hyster-

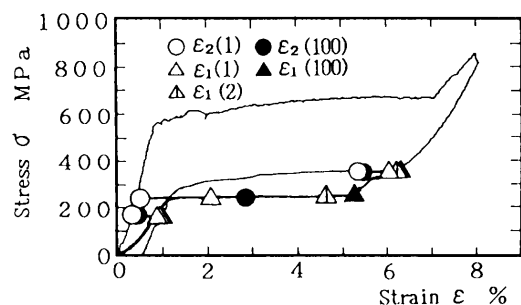


Fig. 8 Stress-strain curves in test with thermal cycling under constant stress ($T_2=373\text{K}$, $\varepsilon_m=1, 2$ and 6%)

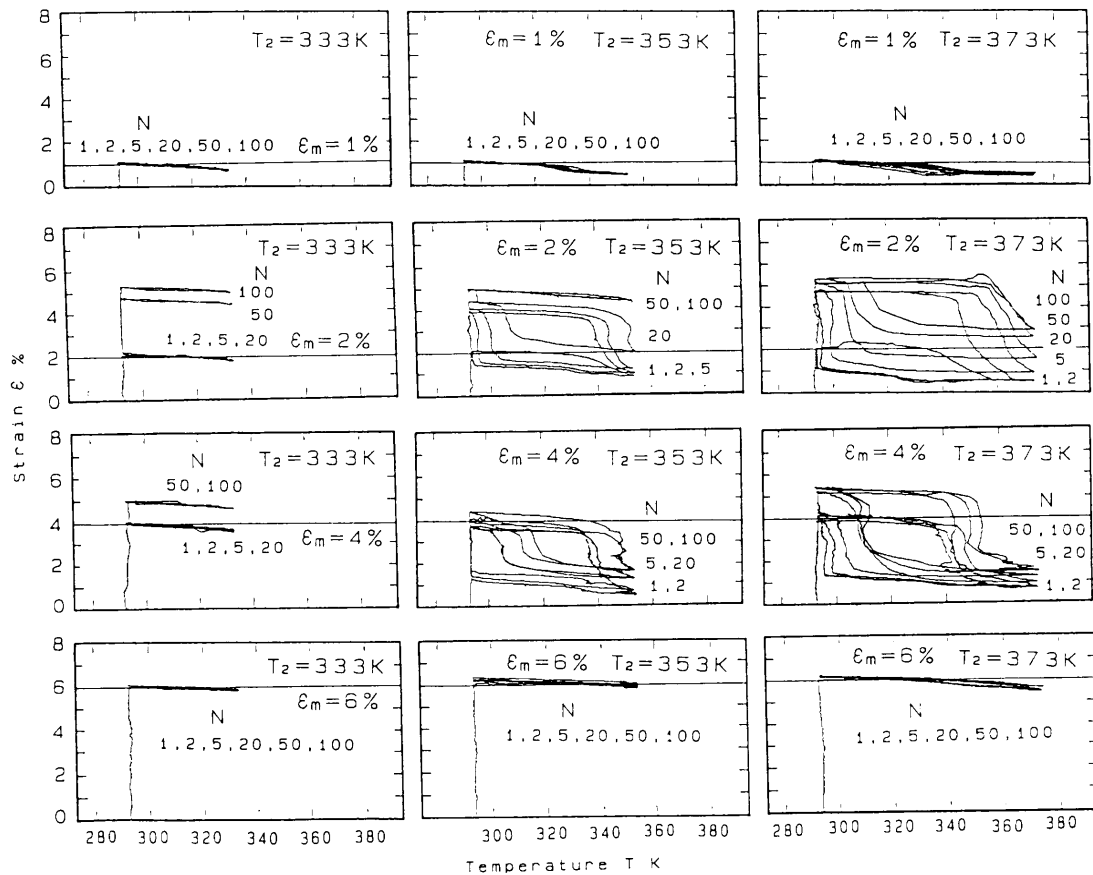


Fig. 9 Strain-temperature curves in test with thermal cycling under constant stress

esis loop of the strain-temperature curve moves toward the right. (3) Both starting strain and completing strain of the SIM transformation and the reverse transformation increase with N , and the position of the hysteresis loop of the strain-temperature curve moves upward. (4) In the case of 2% and 4% for ϵ_m , which are in the SIM transformation region, the strain at low temperature ϵ_1 with thermal cycling becomes 5%, which is equal to the completing strain of the SIM transformation. If T_2 is low, the reverse transformation becomes unapt to appear with thermal cycling, and the hysteresis loop diminishes. (5) In the case of $\epsilon_m = 1\%$, the R-phase transformation occurs stably for thermal cycling. In the case of $\epsilon_m \geq 2\%$, the R-phase transformation disappears with the number of cycles.

4. Conclusions

The cyclic behavior of recovery stress under constant residual strain and recovery strain under constant maximum stress in TiNi shape memory alloy was investigated experimentally. The influence of residual strain, maximum stress and heating temperature under thermal cycling on the cyclic characteristics of recovery stress and recovery strain was

discussed. The main results are summarized as follows.

(1) The stress at high temperature, the stress at low temperature and the recovery stress decrease until the early 30th cycle but become almost constant after these cycles.

(2) The stress-temperature curve moves toward the low-stress direction with the increase in thermal cycling and exhibits a long and thin S-shaped form.

(3) Because the recovery stress under thermal cycling decreases with the increase in residual strain, small residual strain is effective in obtaining a large recovery stress.

(4) Both the stress at high temperature and the recovery stress increase with the rise in the heating temperature, but the rate of increase in both stresses decreases above $A_f + 30$ K.

(5) The amount of decrease in the stress at high temperature under thermal cycling is smaller than that of the starting stress of the stress-induced martensitic transformation by cyclic deformation in the pseudoelastic region.

(6) The transformation temperature under constant stress rises with the number of thermal cycles.

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